Unit 3 - Session 7

SLO 2: GHT for Ellipses & Object Location - Quantify pose estimation accuracy under affine transformations

Duration: 10 Minutes Activity

Question:

The GHT accumulator is a multi-dimensional voting space. Each dimension corresponds to one ellipse parameter: cx (centre x), cy (centre y), a (semi-major axis), b (semi-minor axis), and theta (orientation). When an ellipse is detected, boundary points cast votes and a PEAK forms at the correct parameter location.

When the object undergoes an affine transformation (translation, scaling, rotation, shear), the peak SHIFTS in specific dimensions of the accumulator. Understanding which dimensions are affected — and by how much — is the core of pose estimation accuracy analysis.

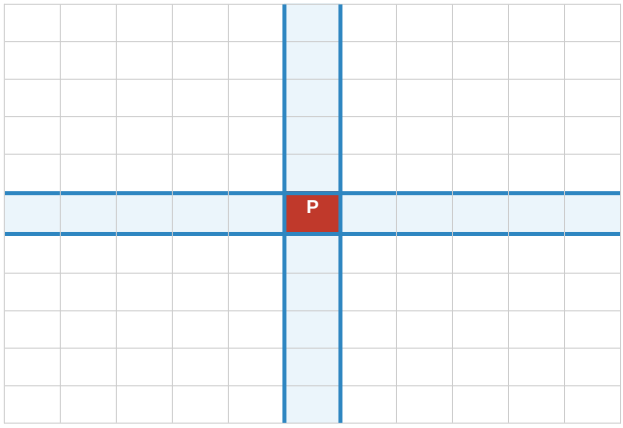
Reference ellipse for this activity: `cx = 5, cy = 5, a = 3, b = 2, theta = 0 deg (peak marked as P on each grid)`

PART A — Reference Peak (cx = 50, cy = 50)

The grid below shows the cx vs cy slice of the accumulator. The reference peak P is at column 5, row 5 (representing cx = 50 px, cy = 50 px). Each cell = 10 px bin. Blue lines mark the accumulator origin axes.

cx vs cy Accumulator Slice

(each cell = 10 px, axis range: 0-100 px)



← cx axis (0 to 100 px) →

REFERENCE PEAK PARAMETERS

`cx = 50 px` *(grid col 5)*
`cy = 50 px` *(grid row 5)*
`a = 30 px` *(semi-major)*
`b = 20 px` *(semi-minor)*
`θ = 0 deg` *(orientation)*

Peak marked as **P** in the grid.

Blue lines = accumulator axes (cx=0 and cy=0).
Each grid cell represents a 10 px bin.

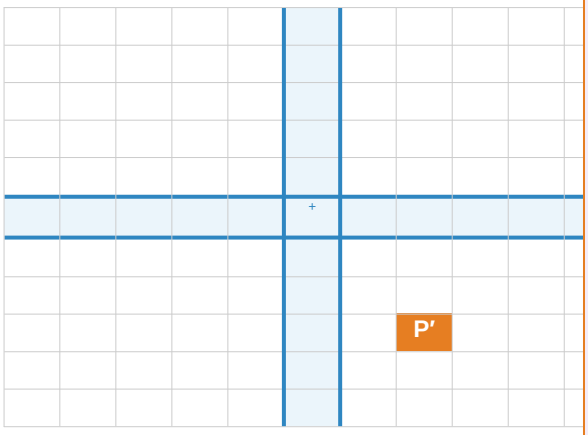
PART B — Apply Each Transformation & Sketch the New Peak

For each transformation: (1) calculate the new parameter values, (2) determine the new grid position, (3) mark the new peak P? on the sketch grid, and (4) fill in the answer fields.

T1 — Translation +20 px in x, +30 px in y

The entire object moves 20 px right and 30 px down in image space.

SKETCH: Draw the new peak position (mark as P?)



← cx axis (0 to 100 px) →

CALCULATE NEW PEAK LOCATION

```
New cx = 50 + 20 = 70 px  (col 7)   |
New cy = 50 + 30 = 80 px  (row 8)
```

New peak at grid position: (7 , 8)

Affected dimensions: `cx, cy`

Unaffected dimensions: `a, b, theta`

Why does the peak shift here?

Reason: Translation moves the object +20 px in x and +30 px in y, so only cx and cy shift.

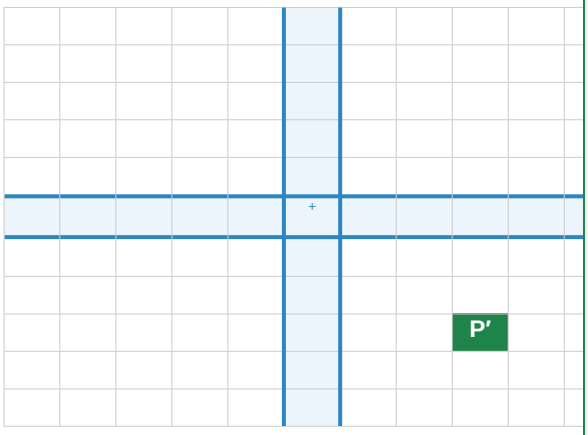
New parameters: cx = 70 px, cy = 80 px, a = 30 px, b = 20 px, theta = 0 deg.

[Hint] *Translation shifts the peak in the cx-cy slice only. Shape parameters are unchanged.*

T2 — Uniform Scaling x 1.5 (about image origin)

The object and its position are scaled up by factor 1.5.

SKETCH: Draw the new peak position (mark as P?)



← cx axis (0 to 100 px) →

CALCULATE NEW PEAK LOCATION

```
New cx = 50 x 1.5 = 75 px  (col 7.5 ~ 8)
|   New cy = 50 x 1.5 = 75 px  (row 7.5 ~
8)
```

New peak at grid position: (8 , 8)

Affected dimensions: `cx, cy, a, b`

Unaffected dimensions: `theta`

Why does the peak shift here?

Reason: Scaling by 1.5 multiplies the centre coordinates and both semi-axes.

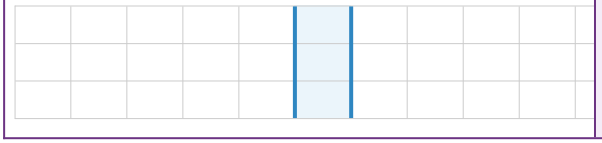
New parameters: cx = 75 px, cy = 75 px, a = 45 px, b = 30 px, theta = 0 deg.

[Hint] *Scaling shifts the peak in cx-cy AND changes a and b proportionally. Orientation is preserved.*

T3 — Rotation 45 degrees (about image centre)

The object rotates 45 degrees counter-clockwise around the image centre (50, 50).

SKETCH: Draw the new peak position (mark as P?)

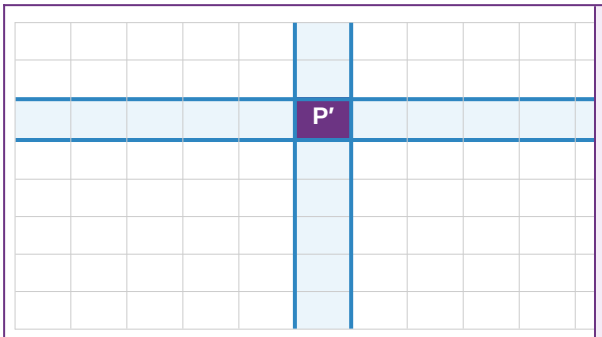


CALCULATE NEW PEAK LOCATION

```
cx, cy shift due to rotation about centre
|   theta changes by +45 deg
```

New peak at grid position: (5 , 5)

Affected dimensions: `cx, cy, theta`



Unaffected dimensions: a, b

Why does the peak shift here?

Reason: Rotation about the ellipse/image centre (50,50) leaves cx and cy unchanged and increases theta by 45 deg.

New parameters: cx = 50 px, cy = 50 px, a = 30 px, b = 20 px, theta = 45 deg.

[Hint] *Rotation about the ellipse centre leaves cx, cy unchanged but shifts theta. Rotation about another point also moves cx, cy.*

PART C — Summary Comparison Table

Complete the table by marking YES or NO for each parameter dimension. Use your answers from Part B. (* = only if rotation is about a point other than the ellipse centre)

Transform	cx shifts?	cy shifts?	a changes?	b changes?	θ changes?	Pose DOF affected
T1 Translation	YES	YES	NO	NO	NO	2 (position)
T2 Scaling x1.5	YES	YES	YES	YES	NO	4 (pos + size)
T3 Rotation 45°	YES*	YES*	NO	NO	YES	3 (pos + orient)
Combined T1+T2	YES	YES	YES	YES	NO	4
Combined T1+T3	YES	YES	NO	NO	YES	3